

2023
(Session : 2022-25)

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the Sections as directed.*

Section – A

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$

(a) Number of power set of $\{a, b\}$, where a and b
are distinct elements :

(i) 3

(ii) 4

(iii) 2

(iv) 5

(b) Which of the following is subset of set $\{1, 2, 3, 4\}$?

(i) $\{1, 2\}$

(ii) $\{1, 2, 3\}$

(iii) $\{1\}$

~~(iv)~~ All of these

(c) $(A \vee \neg A) \vee (q \vee T)$ is a _____.

(i) Tautology

(ii) Contradiction

(iii) Contingency

(iv) None of these

(d) $A \rightarrow (A \vee q)$ is a _____.

(i) Tautology

(ii) Contradiction

(iii) Contingency

(iv) None of these

(e) What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b\}$?

(i) $\{(1, a), (1, b), (2, a), (b, b)\}$

(ii) $\{(1, 1), (2, 2), (a, a), (b, b)\}$

~~(iii) $\{(1, a), (2, a), (1, b), (2, b)\}$~~

(iv) $\{(1, 1), (a, a), (2, a), (1, b)\}$

(f) A function is said to be _____ if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .

(i) One-to-many

~~(ii) One-to-one~~

(iii) Many-to-many

(iv) Many-to-one

(g) An n -vertex graph has _____ edges.

(i) n^2

~~(ii) $n - 1$~~

(iii) $n * n$

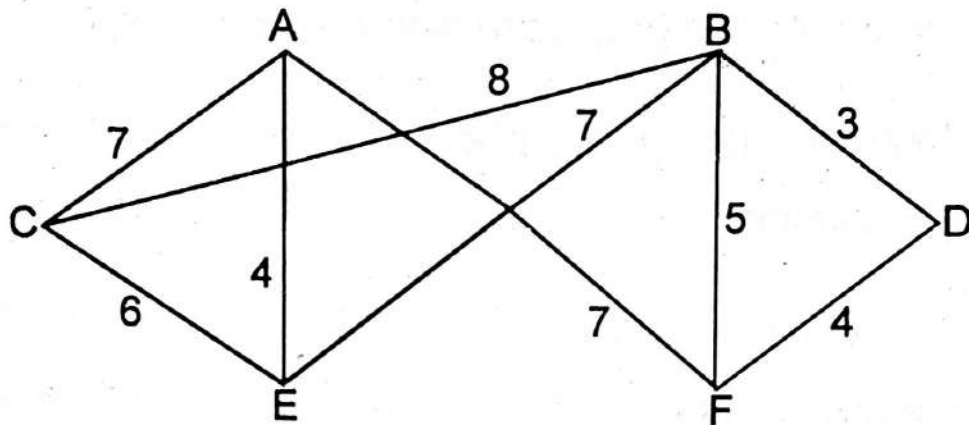
(iv) $n * (n + 1) / 2$

Section – C

Answer any **two** questions of the following :

$$15 \times 2 = 30$$

10. Explain the types of functions.
11. Explain the types of relations.
- ✓12. Find out MST using Krushkal's algorithm :



- ✓13. Using substitution method solve the recurrence relation $T(n) = 2T(n/2) + n$, $T(1) = 1$.

